

Standardizing the Agentic Web: The Agentic Resource Discovery (ARD) Specification

As AI agents evolve from isolated chatbots into autonomous systems that perform complex workflows, a critical bottleneck has emerged: fragmentation. Agents increasingly rely on tools, skills, and other specialized agents distributed across different organizations and platforms. However, until now, there has been no standardized way for an agent to answer three fundamental questions: Where does the right capability live? Which one should I use? And how do I verify it is safe?

The Missing Layer of the Agentic Web

To solve this, Google has introduced the **Agentic Resource Discovery (ARD)** specification. Developed in collaboration with industry partners, ARD is an open standard for publishing, discovering, and verifying AI capabilities across the web. It allows tools and services to be securely shared and connected regardless of their underlying framework or provider.

ARD relies on two key primitives:

- **Catalogs:** Organizations publish an `ai-catalog.json` file on their domain, acting as a verifiable manifest of their available capabilities (such as MCP servers, OpenAPI tools, or other agents).
- **Registries:** These act as search engines for the agentic web, indexing federated catalogs. When an agent queries a registry, it receives matching capabilities along with the necessary trust metadata to establish a secure, direct connection using the tool's native protocol.

Bringing ARD to the Enterprise

Google Cloud is integrating ARD natively into the **Gemini Enterprise Agent Platform** via the Agent Registry. This provides a fully hosted solution for enterprises to govern and operationalize AI resources at scale. It enables organizations to enforce egress policies, pin specific tool versions, and securely verify agent identities to meet stringent compliance standards like HIPAA.

The agent ecosystem thrives on decentralization and openness. By standardizing discovery and trust, ARD provides the critical infrastructure needed to scale autonomous AI systems across organizational boundaries.

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